R2V-Agent: Risk-Calibrated Routing with Robust Verifier-Distillation for Efficient Agentic Systems

Abstract

While scaling test-time compute via Process Reward Models (PRMs) has emerged as a powerful paradigm for enhancing Large Language Model (LLM) reasoning, uniformly allocating expensive search or API calls across every step of a sequential task is highly inefficient. Furthermore, existing LLM routing frameworks are predominantly "stateless," acting at the query level rather than dynamically adapting to the stochasticity of interactive agentic environments (POMDPs). To address this, we introduce R2V-Agent: a framework for step-level Risk-Calibrated Routing coupled with a robust Verifier-Distillation objective. Our approach dynamically allocates test-time compute by continuously evaluating a trajectory's risk using a lightweight Process Verifier. Crucially, we eschew simple entropy heuristics in favor of a theoretical framework constrained by the Conditional Value at Risk (CVaR), providing worst-case robustness bounds against environmental perturbations like tool flakiness and prompt injections. To narrow the intrinsic capability gap, we concurrently train the base Small Language Model (SLM) via a novel Verifier-Guided Direct Preference Optimization (DPO) on deliberately perturbed trajectories. This offline procedure successfully teaches the SLM the "recovery manifold" without the instabilities of multi-turn Reinforcement Learning. Evaluations on complex agentic benchmarks demonstrate that our system achieves frontier-LLM success rates while routing over 85% of steps to the SLM, drastically shifting the cost-efficiency Pareto frontier.

1 Introduction

The deployment of autonomous agents for real-world interactive tasks—such as software engineering, web navigation, and system administration—has been catalyzed by the profound planning capabilities of Large Language Models (LLMs) (Zhou et al., 2024; Mialon et al., 2023). Recently, a pivotal paradigm shift has demonstrated that scaling inference-time compute via Process-based Reward Models (PRMs) and tree-search can be significantly more effective than merely scaling model parameters (Snell et al., 2025; Lightman et al., 2023). However, applying massive test-time compute uniformly across every step is economically intractable. This latency and API cost bottleneck naturally motivates a cascade architecture where a local, computationally efficient Small Language Model (SLM) handles the majority of execution, falling back to a frontier LLM only when necessary (Ong et al., 2025; Chen et al., 2024a; Cai et al., 2024).

Despite the appeal of SLM-LLM cascades, current routing frameworks exhibit critical gaps when applied to sequential decision-making. Existing methods are fundamentally "stateless"; they evaluate a query's difficulty prior to execution and route the entire trajectory accordingly (Ong et al., 2025; Ding et al., 2024). In unstructured Partially Observable Markov Decision Processes (POMDPs), however, difficulty fluctuates mid-trajectory due to stochastic environmental states—such as unexpected tool flakiness, corrupted logs, or partial observability. An SLM making a minor error in step two can suffer from catastrophic compounding errors by step ten. Furthermore, equipping SLMs with self-correction capabilities to survive these stochastic states is notoriously difficult; recent studies demonstrate that standard supervised fine-tuning fails to teach recovery, often necessitating unstable multi-turn Reinforcement Learning (Kumar et al., 2024; Chen et al., 2024b).

To bridge this divide, we propose a mathematical framework for *Risk-Calibrated Allocation of Test-Time Compute*. Rather than routing at the query level, we introduce a step-level trajectory router. At each timestep, a lightweight Process Verifier assesses the downstream risk of the SLM's proposed action (Zhang et al., 2024). To ensure rigorous defense against partial observability and stochastic noise, we formalize the routing objective as a Lagrangian constrained by the Conditional Value at Risk (CVaR) over heterogeneous environmental perturbations (Chow & Ghavamzadeh, 2014; Zhang et al., 2025). This provides formal worst-case bounds on robust task success, dynamically escalating control to a teacher LLM strictly when the system strays outside the SLM's reliable envelope.

To complement our robust routing, we elevate the SLM's intrinsic resilience through *Verifier-Guided DPO on Perturbed Topologies*. By purposefully injecting environmental noise (e.g., tool failures, prompt injections) during offline data generation and utilizing the verifier to score recovery versus failure paths, we bypass the need for full on-policy RL. This structurally distills the LLM's recovery tactics into the SLM using Direct Preference Optimization (Rafailov et al., 2023; Chen et al., 2024c), teaching it the "recovery manifold" in a highly stable manner.

Our contributions are threefold:

- We introduce a novel step-level CVaR routing mechanism that guarantees tail-end performance under stochastic tool noise, drastically outperforming stateless routing baselines.
- We propose a verifier-driven preference distillation strategy that offline-trains SLMs to self-correct in perturbed states without unstable RL.
- We demonstrate through comprehensive evaluations that our risk-calibrated hybrid system preserves the overarching task success rate of monolithic frontier models while utilizing a fraction of the compute.

2 Preliminaries

We model the agent–environment interaction as a Partially Observable Markov Decision Process (POMDP) augmented with a latent perturbation seed,

$$\mathcal{M}_z = (\mathcal{S}, \mathcal{A}, \mathcal{O}, \mathcal{Z}, \mathcal{T}, \mathcal{O}_z, R, \gamma),$$

where $z \in \mathcal{Z}$ captures unobserved stochastic factors such as tool flakiness, latency spikes, truncated logs, or prompt corruption. At step t, the agent observes $o_t \in \mathcal{O}$, selects a tool action $a_t \in \mathcal{A}$, and receives the next observation according to

$$\mathcal{O}_z : \mathcal{S} \times \mathcal{A} \times \mathcal{Z} \to \Delta(\mathcal{O}).$$

We write the agent context as

$$x_t = (G, o_{\leq t}, a_{<t}),$$

where G denotes the task goal. A monolithic frontier policy can be written as $\pi(a \mid x)$, but running such a policy at every step is costly in both latency and API usage. Conversely, a small local model is efficient but often brittle under out-of-distribution perturbations. Our goal is therefore to combine the efficiency of a base SLM with the robustness of a stronger teacher LLM, while exposing the interface between them explicitly enough to permit both theoretical analysis and practical control of worst-case failures.

3 Risk-Calibrated Routing & Verifier-Distillation (R2V)

R2V-Agent factorizes robust agent execution into three components: an efficient base policy $\pi_\theta(a \mid x)$, an expensive teacher policy $\pi_T(a \mid x)$, and a lightweight Process Verifier $V_\phi(x, a) \in [0, 1]$ that scores the quality of candidate actions. A fourth component, the router $r_\psi(f_t) \in [0, 1]$, determines whether the current step should be handled by the SLM or escalated to the teacher.

Algorithm 1 R2V-Agent Training Pipeline

Require: Teacher LLM π_T, initial SLM π_θ, verifier V_ϕ, perturbation operators $\{\mathcal{P}_k\}$
Collect expert trajectories $\mathcal{D}_{\text{exp}}$ using π_T on training tasks
Warm-start the SLM by minimizing $\mathcal{L}_{\text{BC}}$ on $\mathcal{D}_{\text{exp}}$, yielding π_θ^{BC}
Initialize empty datasets $\mathcal{D}_{\text{pref}}, \mathcal{D}_{\text{cons}}, \mathcal{D}_{\text{route}}$
for teacher trajectories and sampled perturbation seeds z **do**
  Replay or branch trajectories with perturbation operators $\mathcal{P}_k$ to obtain perturbed contexts x'_t
  Sample K candidate actions $a_t^{(1:K)} \sim \pi_\theta^{\text{BC}}(\cdot \mid x'_t)$
  Score candidates with $V_\phi(x'_t, a_t^{(k)})$; optionally query teacher action $a_t^T \sim \pi_T(\cdot \mid x'_t)$
  Construct preference pairs (x'_t, a_t^+, a_t^-) and add them to $\mathcal{D}_{\text{pref}}$
  Construct paired perturbation views for consistency regularization and add them to $\mathcal{D}_{\text{cons}}$
end for
Continue training the SLM from π_θ^{BC} using $\mathcal{L}_{\text{BC}} + \beta_1 \mathcal{L}_{\text{DPO}} + \beta_2 \mathcal{L}_{\text{cons}}$, yielding distilled policy π_θ
for rollouts of the distilled policy π_θ under diverse perturbation seeds **do**
  At each step, sample SLM candidates, score them with V_ϕ, and extract risk features f_t
  Define escalation label y_t indicating whether teacher intervention would have been beneficial
  Add (f_t, y_t) to $\mathcal{D}_{\text{route}}$
end for
Train router r_ψ on $\mathcal{D}_{\text{route}}$ with Brier calibration and a CVaR-constrained cost objective
return Distilled SLM π_θ, verifier V_ϕ, router r_ψ

The key design choice is that these components are *trained sequentially*, not jointly. We first warm-start the SLM with behavioral cloning on expert demonstrations. We then use this BC policy to collect perturbed rollouts, from which the verifier constructs preference pairs and consistency pairs that further train the SLM to recover under stochastic perturbations. Only after this distilled policy is fixed do we train the router, whose job is no longer to compensate for a weak base model from scratch, but to recognize the residual high-risk states where teacher intervention is still necessary. This ordering is important: the router is trained on the actual failure modes of the post-distillation SLM, and thus calibrates to the policy that will be deployed.

The following assumptions are used only for the formal guarantees below.

Assumption 1 (Boundedness and measurability)**.** *For every step t,*

$$V_\phi(x_t, a_t) \in [0,1], \qquad r_\psi(f_t) \in [0,1], \qquad y_t \in \{0,1\}, \qquad S_z \in \{0,1\}.$$

Moreover, the feature vector satisfies

$$\|f_t\|_2 \le B_f$$

for some finite constant $B_f > 0$.

Assumption 2 (Well-defined calibration target)**.** *Define the conditional escalation probability*

$$q^\star(f) := \mathbb{P}(y_t = 1 \mid f_t = f).$$

We assume that $q^\star(f)$ is well-defined for almost every f.

3.1 Verifier-Guided Distillation on Perturbed Topologies

We begin by training a competent *base* SLM before introducing any routing. Let $\mathcal{D}_{\text{exp}} = \{(x_t, a_t^\star)\}$ denote expert demonstrations collected from the teacher policy π_T. The initial policy is obtained via standard behavioral cloning:

$$\mathcal{L}_{\text{BC}}(\theta) = -\mathbb{E}_{(x_t, a_t^\star) \sim \mathcal{D}_{\text{exp}}} \left[\log \pi_\theta(a_t^\star \mid x_t) \right].$$

This BC stage is essential in practice: it gives the SLM a competent nominal policy before we ask it to learn recovery under perturbation.

Starting from this warm-started policy π_θ^{BC}, we then construct a *perturbed recovery dataset*. Specifically, we apply perturbation operators $\mathcal{P}_k$ to replayed or branched trajectories, generating out-of-distribution contexts x_t' that mimic tool failures, missing observations, adversarial strings, or other stochastic disruptions. At each such context, the BC model proposes candidate actions

$$a_t^{(1)}, \ldots, a_t^{(K)} \sim \pi_\theta^{\text{BC}}(\cdot \mid x_t'),$$

which are scored by the verifier V_ϕ. When necessary, we also query the teacher policy to obtain a recovery action a_t^T. These candidates are then converted into preference pairs (a_t^+, a_t^-), where a_t^+ is a verifier-approved or teacher-recovered action and a_t^- is a lower-quality alternative sampled from the BC policy. In this way, the DPO stage is trained on *the failure modes of the BC model itself*, rather than on generic preference data detached from deployment.

Let

$$\mathcal{G}_\gamma(x_t) = \{a \in \mathcal{A} : V_\phi(x_t, a) \geq \gamma\}$$

denote the set of verifier-good actions at threshold γ. The next proposition formalizes why sampling multiple SLM candidates and selecting with the verifier increases the probability of recovering a good action.

Proposition 3 (Best-of-K verifier selection improves with coverage)**.** *Fix a context x_t, and suppose K candidate actions are sampled i.i.d. from $\pi_\theta(\cdot \mid x_t)$. Assume that for every good action $a^+ \in \mathcal{G}_\gamma(x_t)$ and every bad action $a^- \notin \mathcal{G}_\gamma(x_t)$, the verifier misranks the pair with probability at most η_V, i.e.,*

$$\mathbb{P}\big(V_\phi(x_t, a^-) \geq V_\phi(x_t, a^+)\big) \leq \eta_V.$$

Then the probability that verifier selection returns a γ-good action satisfies

$$\mathbb{P}\Big(\hat{a}_t^{\text{SLM}} \in \mathcal{G}_\gamma(x_t)\Big) \geq 1 - (1 - \mu_\gamma(x_t))^K - \frac{K(K-1)}{2}\eta_V.$$

(Proof deferred to Appendix A)

We then continue training the SLM with Direct Preference Optimization on these perturbed states:

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(x_t', a^+, a^-)}\left[\log \sigma\left(\beta\left[\log \frac{\pi_\theta(a^+ \mid x_t')}{\pi_{\text{ref}}(a^+ \mid x_t')} - \log \frac{\pi_\theta(a^- \mid x_t')}{\pi_{\text{ref}}(a^- \mid x_t')}\right]\right)\right].$$

This objective explicitly pushes the SLM toward actions that recover from perturbations, but avoids unstable multi-turn RL by training entirely offline on verifier-constructed preference data.

Proposition 4 (Pairwise sign consistency of verifier-noisy DPO)**.** *Consider a fixed context x_t and two actions a^+, a^-. Let the true pairwise preference label be*

$$y^\star \in \{+1, -1\},$$

where $y^\star = +1$ means a^+ is truly preferred to a^-. Suppose the verifier produces a noisy pairwise label $\widetilde{y}$ such that

$$\mathbb{P}(\widetilde{y} = y^\star \mid x_t, a^+, a^-) = 1 - \eta, \qquad \mathbb{P}(\widetilde{y} = -y^\star \mid x_t, a^+, a^-) = \eta,$$

with $\eta < 1/2$.

Let the DPO score difference be

$$u_\theta(x_t, a^+, a^-) = \beta\left[\log \frac{\pi_\theta(a^+ \mid x_t)}{\pi_{\text{ref}}(a^+ \mid x_t)} - \log \frac{\pi_\theta(a^- \mid x_t)}{\pi_{\text{ref}}(a^- \mid x_t)}\right].$$

Then the population minimizer of the noisy logistic loss

$$\mathcal{L}_{\text{DPO,noisy}}(\theta) = \mathbb{E}\Big[\log(1 + \exp(-\widetilde{y}\, u_\theta))\Big]$$

has the same sign as the true preference:

$$\operatorname{sign}\big(u_\theta^\star(x_t, a^+, a^-)\big) = y^\star.$$

(Proof deferred to Appendix A)

To further stabilize the policy under partial observability, we regularize the SLM to behave consistently across different perturbation realizations of the same latent state. Concretely, if two perturbation seeds z and z' produce two observed contexts corresponding to the same latent state, we penalize divergence between the induced action distributions:

$$\mathcal{L}_{\text{cons}}(\theta) = \mathbb{E}_{t,z,z'}\left[JSD\Big(\pi_\theta(\cdot \mid x_t^{(z)}) \;\Big\|\; \pi_\theta(\cdot \mid x_t^{(z')})\Big)\right].$$

Assumption 5 (Perturbation pairing and stepwise Lipschitzness)**.** *Suppose two perturbation seeds $z, z' \in \mathcal{Z}$ produce two observed contexts $x_t^{(z)}$ and $x_t^{(z')}$ corresponding to the same latent environment state s_t. Let $P_t^{(z)} := \pi_\theta(\cdot \mid x_t^{(z)})$ and $P_t^{(z')} := \pi_\theta(\cdot \mid x_t^{(z')})$. Assume the one-step surrogate loss $\ell_t(P; s_t) \in [0, 1]$ satisfies*

$$|\ell_t(P; s_t) - \ell_t(Q; s_t)| \le L\,(P, Q)$$

for all action distributions P, Q and some constant $L > 0$.

Lemma 6 (Total variation is controlled by Jensen–Shannon divergence)**.** *For any two distributions P and Q,*

$$(P, Q) \le \sqrt{2\,(P\|Q)}.$$

(Proof deferred to Appendix A)

Theorem 7 (Perturbation-generalization bound from consistency regularization)**.** *Assume Assumptions 14 and 17. For a perturbation seed z, define the episodic surrogate risk*

$$R_z(\pi_\theta) := \mathbb{E}\left[\sum_{t=1}^{H} \ell_t\Big(\pi_\theta(\cdot \mid x_t^{(z)}); s_t\Big)\right].$$

Then for any two perturbation seeds $z, z' \in \mathcal{Z}$,

$$|R_z(\pi_\theta) - R_{z'}(\pi_\theta)| \le HL\sqrt{2\,\mathcal{L}_{\text{cons}}(\theta)},$$

where

$$\mathcal{L}_{\text{cons}}(\theta) = \mathbb{E}_{t,z,z'}\left[\Big(\pi_\theta(\cdot \mid x_t^{(z)}) \;\Big\|\; \pi_\theta(\cdot \mid x_t^{(z')})\Big)\right].$$

Consequently, if a training seed z_{train} is representative of the seen perturbations, then for any unseen test seed z_{test},

$$R_{z_{\text{test}}}(\pi_\theta) \le R_{z_{\text{train}}}(\pi_\theta) + HL\sqrt{2\,\mathcal{L}_{\text{cons}}(\theta)}.$$

(Proof deferred to Appendix A)

The distilled SLM is therefore obtained in two explicit stages: BC first, then perturbation-aware preference distillation. The final policy objective is

$$\min_\theta \quad \mathcal{L}_{\text{BC}} + \beta_1 \mathcal{L}_{\text{DPO}} + \beta_2 \mathcal{L}_{\text{cons}},$$

where optimization starts from the BC warm-start rather than from a random initialization.

3.2 Risk-Calibrated Trajectory Routing via CVaR

Once the distilled SLM π_θ is fixed, we train a step-level router that decides whether the next action should be executed by the SLM or escalated to the teacher LLM. This ordering matters: the router should model the *residual risk* of the already-distilled policy, not the behavior of an earlier, weaker SLM.

For each context x_t, the distilled SLM proposes K candidate actions. The verifier scores these candidates, and we summarize the resulting uncertainty into a compact feature vector

$$f_t = \Phi(x_t, \pi_\theta, V_\phi),$$

which may include the policy entropy, the maximum/mean/standard deviation of verifier scores, recent verifier trends, tool-failure indicators, and other execution statistics. We then assign a binary escalation label $y_t \in \{0, 1\}$, where $y_t = 1$ indicates that the current step is one on which escalation to the teacher would have been beneficial. In the simplest instantiation, this label is obtained by comparing the best verifier-scored SLM candidate against the teacher action on the same context.

The router $r_\psi(f_t) \in [0, 1]$ is trained as a calibrated estimate of this escalation probability. At deployment, it is thresholded to produce a hard routing decision $d_t \in \{0, 1\}$, which determines whether the system uses the local SLM or spends teacher compute on the current step.

Assumption 8 (Cost-sensitive routing surrogate)**.** *For theoretical analysis of the hard router, we measure the quality of a routing decision $d_t \in \{0, 1\}$ using the stepwise surrogate loss*

$$\ell_{\text{route}}(d_t, y_t) = c_{\text{SLM}}(1 - d_t) + c_{\text{LLM}} d_t + \kappa\, y_t(1 - d_t),$$

where $c_{\text{SLM}} < c_{\text{LLM}}$ and $\kappa > 0$ is the penalty for not *escalating on a step where escalation was actually needed.*

We optimize the router with a cost-sensitive objective regularized by both calibration and worst-case robustness. Let $S(z) \in \{0, 1\}$ denote terminal success under perturbation seed z. To control heavy-tail failures, we constrain the Conditional Value at Risk of the failure random variable $1 - S(z)$:

$$\min_{\psi} \max_{\lambda \geq 0} \; \mathbb{E}_{x \sim \mathcal{D}}\big[c_{\text{LLM}} \cdot d_t + c_{\text{SLM}} \cdot (1 - d_t)\big] + \lambda \Big(CVaR_\alpha(1 - S(z)) - \epsilon \Big) + \lambda_B \mathcal{L}_{\text{Brier}}(\psi).$$

The Brier term calibrates the router probabilities, while the CVaR term discourages routing policies that appear cheap on average but fail catastrophically on the worst perturbation seeds.

Theorem 9 (Brier-optimal calibration of the router)**.** *Under Assumptions 14 and 15, the population minimizer of the Brier risk*

$$\mathcal{R}_{\text{Brier}}(r_\psi) = \mathbb{E}\big[(r_\psi(f_t) - y_t)^2\big]$$

is the conditional escalation probability

$$r^\star(f) = q^\star(f) = \mathbb{P}(y_t = 1 \mid f_t = f).$$

Equivalently, among all measurable router outputs $p(f) \in [0, 1]$, the unique pointwise minimizer is the true escalation probability.

(Proof deferred to Appendix A)

Corollary 10 (Cost-optimal hard routing threshold)**.** *Under Assumptions 14, 15, and 16, the Bayes-optimal hard routing rule for the stepwise loss ℓ_{route} is*

$$d_t^\star = 1\left[q^\star(f_t) \geq \frac{c_{\text{LLM}} - c_{\text{SLM}}}{\kappa}\right].$$

Equivalently, if the router is calibrated so that $r_\psi(f_t) \approx q^\star(f_t)$, then the optimal escalation rule is to compare the router probability against the threshold

$$\tau^\star_{\text{route}} = \frac{c_{\text{LLM}} - c_{\text{SLM}}}{\kappa}.$$

(Proof deferred to Appendix A)

Assumption 11 (Convex surrogate for primal–dual analysis)**.** *For the convergence theorem only, we analyze the convex surrogate router*

$$p_\psi(f) = \sigma(\psi^\top f / T),$$

where $T > 0$ is fixed and ψ lies in a compact convex set Ψ. Let

$$g(\psi) := \widehat{\ }_\alpha(1 - S_z) - \varepsilon$$

denote the empirical CVaR-constraint violation. Assume that the empirical Lagrangian

$$\widehat{\mathcal{L}}_{\text{router}}(\psi, \lambda) = \widehat{\mathbb{E}}[\text{cost}(d)] + \lambda\, g(\psi) + \widehat{\ }(r_\psi(f), y)$$

is convex in ψ, concave in λ, and has gradients bounded by G over $\Psi \times [0, \Lambda]$.

Algorithm 2 R2V-Agent Risk-Calibrated Inference Loop

Require: Distilled SLM π_θ, Teacher LLM π_T, Verifier V_ϕ, Router r_ψ, Threshold $\tau^\star_{\text{route}}$

Initialize context $x_0 = (G, o_0)$
for $t = 0, 1, \ldots, H-1$ **do**
  Sample K candidate actions $a_t^{(1)}, \ldots, a_t^{(K)} \sim \pi_\theta(\cdot \mid x_t)$
  Score candidates $s_t^{(k)} = V_\phi(x_t, a_t^{(k)})$ and set $\hat{a}_t^{\text{SLM}} = \arg\max_k s_t^{(k)}$
  Compute risk features $f_t = \Phi(x_t, \pi_\theta, V_\phi)$
  Compute escalation probability $p_t = r_\psi(f_t)$
  if $p_t \geq \tau^\star_{\text{route}}$ **and** Remaining Budget > 0 **then**
    Sample $a_t \sim \pi_T(\cdot \mid x_t)$
    Decrement LLM Budget
  else
    Set $a_t = \hat{a}_t^{\text{SLM}}$
  end if
  Execute a_t, observe $o_{t+1} \sim \mathcal{O}_z(\cdot \mid x_t, a_t, z)$
  Update context $x_{t+1} = (x_t, a_t, o_{t+1})$
end for

Theorem 12 (Primal–dual convergence and near-feasibility)**.** *Assume Assumptions 14 and 18. Let the diameter of Ψ be at most D_ψ, and the diameter of $[0, \Lambda]$ be at most $D_\lambda = \Lambda$. Define*

$$D := \sqrt{D_\psi^2 + D_\lambda^2}.$$

If the step size is chosen as

$$\eta = \frac{D}{G\sqrt{N}},$$

then the averaged iterates satisfy the saddle-point gap bound

$$\max_{\lambda \in [0,\Lambda]} \widehat{\mathcal{L}}_{\text{router}}(\bar{\psi}_N, \lambda) - \min_{\psi \in \Psi} \widehat{\mathcal{L}}_{\text{router}}(\psi, \bar{\lambda}_N) \leq \frac{2DG}{\sqrt{N}}.$$

Moreover, the empirical CVaR-constraint violation satisfies

$$\left[\widehat{\mathrm{CVaR}}_\alpha(1 - S_z; \bar{\psi}_N) - \varepsilon\right]_+ \leq \frac{2DG}{\Lambda\sqrt{N}}.$$

(Proof deferred to Appendix A)

Theorem 13 (Regret bound from router miscalibration)**.** *Let*

$$\hat{d}_t = \left[r_\psi(f_t) \geq \frac{c_{\text{LLM}} - c_{\text{SLM}}}{\kappa}\right]$$

be the plug-in hard router obtained from a probabilistic router r_ψ, and let $d_t^\star$ be the Bayes-optimal rule from Corollary 20. Then

$$\mathbb{E}\left[\ell_{\text{route}}(\hat{d}_t, y_t) - \ell_{\text{route}}(d_t^\star, y_t)\right] \leq \kappa\, \mathbb{E}\left[|r_\psi(f_t) - q^\star(f_t)|\right].$$

(Proof deferred to Appendix A)

3.3 Inference with Risk-Calibrated Escalation

At test time, the distilled SLM handles the default path, while the teacher LLM is reserved for high-risk steps identified by the router. The full inference procedure is summarized below.

This inference rule makes the separation of responsibilities explicit: the SLM provides cheap high-throughput execution, the verifier scores local action quality, and the router allocates expensive teacher compute only when the predicted risk exceeds the calibrated threshold.

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A Theoretical Guarantees for Risk-Calibrated Routing

We study the agent defined on the noisy-observation POMDP

$$\mathcal{M}_z = (\mathcal{S}, \mathcal{A}, \mathcal{O}, T, O_z, R, \gamma),$$

where $z \in \mathcal{Z}$ denotes the perturbation seed, x_t denotes the step-t observation context, π_θ is the SLM policy, π_T is the teacher LLM, V_ϕ is the verifier, and r_ψ is the router.

At each step t, the SLM generates K candidates

$$a_t^{(1)}, \ldots, a_t^{(K)} \sim \pi_\theta(\cdot \mid x_t),$$

the verifier scores them by

$$s_t^{(k)} = V_\phi(x_t, a_t^{(k)}),$$

and the best SLM candidate is

$$\hat{a}_t^{\mathrm{SLM}} = \arg\max_{1 \le k \le K} V_\phi(x_t, a_t^{(k)}).$$

The router receives the feature vector $f_t \in \mathbb{R}^{13}$ and outputs

$$p_t = r_\psi(f_t) \in [0, 1].$$

A hard routing decision is then obtained by thresholding:

$$d_t = \mathbf{1}[p_t \ge \tau_{\mathrm{route}}].$$

Throughout the theory section, let $y_t \in \{0, 1\}$ denote the *correct escalation label* at step t. This binary label can be instantiated in either of the two ways already used in the methodology:

$$y_t = \mathbf{1}\left[\max_k V_\phi(x_t, a_{\mathrm{SLM}}^{(k)}) < V_\phi(x_t, a_{\mathrm{teacher}})\right]$$

or

$$y_t = \mathbf{1}[\rho(x_t) > \tau], \qquad \rho(x_t) = f\Big(1 - V_\phi(x_t, \hat{a}_t^{\mathrm{SLM}}),\ H(\pi_\theta(\cdot \mid x_t))\Big).$$

All results below only require that y_t be a valid binary escalation target.

A.1 Assumptions

Assumption 14 (Boundedness and measurability)**.** *For every step t,*

$$V_\phi(x_t, a_t) \in [0, 1], \qquad r_\psi(f_t) \in [0, 1], \qquad y_t \in \{0, 1\}, \qquad S_z \in \{0, 1\}.$$

Moreover, the feature vector satisfies

$$\|f_t\|_2 \le B_f$$

for some finite constant $B_f > 0$.

Assumption 15 (Well-defined calibration target)**.** *Define the conditional escalation probability*

$$q^\star(f) := \mathbb{P}(y_t = 1 \mid f_t = f).$$

We assume that $q^\star(f)$ is well-defined for almost every f.

Assumption 16 (Cost-sensitive routing surrogate)**.** *For theoretical analysis of the hard router, we measure the quality of a routing decision $d_t \in \{0, 1\}$ using the stepwise surrogate loss*

$$\ell_{\mathrm{route}}(d_t, y_t) = c_{\mathrm{SLM}}(1 - d_t) + c_{\mathrm{LLM}} d_t + \kappa\, y_t(1 - d_t),$$

where $c_{\mathrm{SLM}} < c_{\mathrm{LLM}}$ and $\kappa > 0$ is the penalty for not *escalating on a step where escalation was actually needed.*

Assumption 17 (Perturbation pairing and stepwise Lipschitzness)**.** *Suppose two perturbation seeds $z, z' \in \mathcal{Z}$ produce two observed contexts $x_t^{(z)}$ and $x_t^{(z')}$ corresponding to the same latent environment state s_t. Let $P_t^{(z)} := \pi_\theta(\cdot \mid x_t^{(z)})$ and $P_t^{(z')} := \pi_\theta(\cdot \mid x_t^{(z')})$. Assume the one-step surrogate loss $\ell_t(P; s_t) \in [0, 1]$ satisfies*

$$|\ell_t(P; s_t) - \ell_t(Q; s_t)| \leq L \ \mathrm{TV}(P, Q)$$

for all action distributions P, Q and some constant $L > 0$.

Assumption 18 (Convex surrogate for primal–dual analysis)**.** *For the convergence theorem only, we analyze the convex surrogate router*

$$p_\psi(f) = \sigma(\psi^\top f / T),$$

where $T > 0$ is fixed and ψ lies in a compact convex set Ψ. Let

$$g(\psi) := \widehat{\mathrm{CVaR}}_\alpha(1 - S_z) - \varepsilon$$

denote the empirical CVaR-constraint violation. Assume that the empirical Lagrangian

$$\widehat{\mathcal{L}}_{\text{router}}(\psi, \lambda) = \widehat{\mathbb{E}}[\mathrm{cost}(d)] + \lambda \, g(\psi) + \widehat{\mathrm{Brier}}(r_\psi(f), y)$$

is convex in ψ, concave in λ, and has gradients bounded by G over $\Psi \times [0, \Lambda]$.

Remark A.1. *Assumption 18 is the standard place where theory becomes slightly idealized relative to the practical MLP router. The theorem should therefore be presented as justification for the* training principle*—namely CVaR-constrained calibrated routing—rather than as a global convergence guarantee for the full nonconvex MLP.*

A.2 Calibration and Bayes-optimal routing

Theorem 19 (Brier-optimal calibration of the router)**.** *Under Assumptions 14 and 15, the population minimizer of the Brier risk*

$$\mathcal{R}_{\text{Brier}}(r_\psi) = \mathbb{E}\big[(r_\psi(f_t) - y_t)^2\big]$$

is the conditional escalation probability

$$r^\star(f) = q^\star(f) = \mathbb{P}(y_t = 1 \mid f_t = f).$$

Equivalently, among all measurable router outputs $p(f) \in [0, 1]$, the unique pointwise minimizer is the true escalation probability.

Proof. We prove the result by conditioning on the feature vector f_t.

Step 1: Write the population Brier risk using the tower property. By the law of total expectation,

$$\mathcal{R}_{\text{Brier}}(p) = \mathbb{E}\big[(p(f_t) - y_t)^2\big] = \mathbb{E}\Big[\mathbb{E}\big[(p(f_t) - y_t)^2 \mid f_t\big]\Big].$$

Therefore, to minimize the outer expectation, it is enough to minimize the inner conditional quantity for each fixed feature value f.

Step 2: Fix a feature value f and denote $p = p(f)$, $q = q^\star(f)$. By definition,

$$q = \mathbb{P}(y_t = 1 \mid f_t = f).$$

Because $y_t \in \{0, 1\}$, conditional on $f_t = f$, the random variable y_t equals 1 with probability q and 0 with probability $1 - q$.

Hence

$$\mathbb{E}\big[(p - y_t)^2 \mid f_t = f\big] = q(p - 1)^2 + (1 - q)(p - 0)^2.$$

Step 3: Expand the squares carefully. We have

$$(p-1)^2 = p^2 - 2p + 1, \qquad (p-0)^2 = p^2.$$

Substitute these into the conditional risk:

$$q(p^2 - 2p + 1) + (1-q)p^2.$$

Now distribute the coefficients:

$$= qp^2 - 2qp + q + p^2 - qp^2.$$

The two qp^2 terms cancel:

$$= p^2 - 2qp + q.$$

Step 4: Complete the square. Add and subtract q^2:

$$p^2 - 2qp + q = (p^2 - 2qp + q^2) + q - q^2 = (p-q)^2 + q(1-q).$$

Step 5: Identify the minimizing value. The term $q(1-q)$ does not depend on p. The term $(p-q)^2$ is always nonnegative and equals zero if and only if $p = q$.

Therefore, for every fixed f,

$$\arg\min_{p \in [0,1]} \mathbb{E}\left[(p - y_t)^2 \mid f_t = f\right] = q^\star(f).$$

Step 6: Lift the pointwise minimizer back to the population problem. Since the population risk is the expectation of the conditional risk, and the conditional risk is minimized pointwise by $p(f) = q^\star(f)$, the global minimizer is

$$r^\star(f) = q^\star(f).$$

This proves the theorem. □

Corollary 20 (Cost-optimal hard routing threshold)**.** *Under Assumptions 14, 15, and 16, the Bayes-optimal hard routing rule for the stepwise loss ℓ_{route} is*

$$d_t^\star = \mathbf{1}\left[q^\star(f_t) \geq \frac{c_{\text{LLM}} - c_{\text{SLM}}}{\kappa}\right].$$

Equivalently, if the router is calibrated so that $r_\psi(f_t) \approx q^\star(f_t)$, then the optimal escalation rule is to compare the router probability against the threshold

$$\tau_{\text{route}}^\star = \frac{c_{\text{LLM}} - c_{\text{SLM}}}{\kappa}.$$

Proof. Fix a feature value f, and let $q = q^\star(f) = \mathbb{P}(y_t = 1 \mid f_t = f)$.

We compare the conditional expected loss of the two possible actions $d_t = 1$ and $d_t = 0$.

Case 1: Escalate, i.e., $d_t = 1$. Substitute $d_t = 1$ into

$$\ell_{\text{route}}(d_t, y_t) = c_{\text{SLM}}(1 - d_t) + c_{\text{LLM}} d_t + \kappa y_t (1 - d_t).$$

Because $1 - d_t = 0$ and $d_t = 1$, this becomes

$$\ell_{\text{route}}(1, y_t) = c_{\text{LLM}}.$$

Therefore

$$\mathbb{E}[\ell_{\text{route}}(1, y_t) \mid f_t = f] = c_{\text{LLM}}.$$

Case 2: Do not escalate, i.e., $d_t = 0$**.** Now $1 - d_t = 1$ and $d_t = 0$, so

$$\ell_{\text{route}}(0, y_t) = c_{\text{SLM}} + \kappa y_t.$$

Taking conditional expectation,

$$\mathbb{E}[\ell_{\text{route}}(0, y_t) \mid f_t = f] = c_{\text{SLM}} + \kappa \mathbb{E}[y_t \mid f_t = f] = c_{\text{SLM}} + \kappa q.$$

Step 3: Compare the two conditional risks. Escalation is optimal if and only if

$$c_{\text{LLM}} \leq c_{\text{SLM}} + \kappa q.$$

Subtract c_{SLM} from both sides:

$$c_{\text{LLM}} - c_{\text{SLM}} \leq \kappa q.$$

Now divide by $\kappa > 0$:

$$q \geq \frac{c_{\text{LLM}} - c_{\text{SLM}}}{\kappa}.$$

Thus the Bayes-optimal rule is

$$d_t^\star = \mathbf{1}\left[q^\star(f_t) \geq \frac{c_{\text{LLM}} - c_{\text{SLM}}}{\kappa}\right].$$

This proves the corollary. □

Theorem 21 (Regret bound from router miscalibration)**.** *Let*

$$\hat{d}_t = \mathbf{1}\left[r_\psi(f_t) \geq \frac{c_{\text{LLM}} - c_{\text{SLM}}}{\kappa}\right]$$

be the plug-in hard router obtained from a probabilistic router r_ψ*, and let* $d_t^\star$ *be the Bayes-optimal rule from Corollary 20. Then*

$$\mathbb{E}\left[\ell_{\text{route}}(\hat{d}_t, y_t) - \ell_{\text{route}}(d_t^\star, y_t)\right] \leq \kappa\, \mathbb{E}\left[|r_\psi(f_t) - q^\star(f_t)|\right].$$

Proof. We again work conditionally on $f_t = f$. Let

$$q = q^\star(f), \qquad \widehat{q} = r_\psi(f), \qquad \tau^\star = \frac{c_{\text{LLM}} - c_{\text{SLM}}}{\kappa}.$$

Step 1: Compute the conditional risk difference between the two actions. From the previous proof,

$$R_f(1) := \mathbb{E}[\ell_{\text{route}}(1, y_t) \mid f_t = f] = c_{\text{LLM}},$$

and

$$R_f(0) := \mathbb{E}[\ell_{\text{route}}(0, y_t) \mid f_t = f] = c_{\text{SLM}} + \kappa q.$$

Therefore,

$$R_f(0) - R_f(1) = c_{\text{SLM}} + \kappa q - c_{\text{LLM}} = \kappa q - (c_{\text{LLM}} - c_{\text{SLM}}) = \kappa(q - \tau^\star).$$

So the absolute advantage of the better action over the worse action is

$$|R_f(0) - R_f(1)| = \kappa |q - \tau^\star|.$$

Step 2: Characterize when the plug-in router makes a mistake. The optimal decision is

$$d^\star(f) = \mathbf{1}[q \geq \tau^\star],$$

while the plug-in decision is

$$\hat{d}(f) = \mathbf{1}[\widehat{q} \geq \tau^\star].$$

These two decisions differ only if q and $\widehat{q}$ lie on opposite sides of the threshold $\tau^\star$.

Hence, on the event $\hat{d}(f) \neq d^\star(f)$, the threshold $\tau^\star$ lies between q and $\widehat{q}$, so

$$|q - \tau^\star| \leq |\widehat{q} - q|.$$

Step 3: Bound the conditional regret. If $\hat{d}(f) = d^\star(f)$, then the conditional regret is 0.

If $\hat{d}(f) \neq d^\star(f)$, then the conditional regret equals exactly the difference between the worse and better conditional actions, namely

$$\kappa|q - \tau^\star|.$$

Using the previous step,

$$\kappa|q - \tau^\star| \leq \kappa|\widehat{q} - q|.$$

Therefore, in all cases,

$$\mathbb{E}\Big[\ell_{\text{route}}(\hat{d}_t, y_t) - \ell_{\text{route}}(d_t^\star, y_t) \mid f_t = f\Big] \leq \kappa|\widehat{q} - q|.$$

Step 4: Average over f_t. Take expectation over f_t:

$$\mathbb{E}\Big[\ell_{\text{route}}(\hat{d}_t, y_t) - \ell_{\text{route}}(d_t^\star, y_t)\Big] \leq \kappa\mathbb{E}\left[|r_\psi(f_t) - q^\star(f_t)|\right].$$

This proves the theorem. □

A.3 Verifier-guided candidate generation

Definition A.1 (Good candidate set)**.** *For a fixed context x_t and threshold $\gamma \in [0,1]$, define*

$$\mathcal{G}_\gamma(x_t) := \left\{a \in \mathcal{A} : V_\phi(x_t, a) \geq \gamma\right\}.$$

Also define the SLM coverage probability

$$\mu_\gamma(x_t) := \mathbb{P}_{a \sim \pi_\theta(\cdot|x_t)}\big(a \in \mathcal{G}_\gamma(x_t)\big).$$

Proposition 22 (Best-of-K verifier selection improves with coverage)**.** *Fix a context x_t, and suppose K candidate actions are sampled i.i.d. from $\pi_\theta(\cdot \mid x_t)$. Assume that for every good action $a^+ \in \mathcal{G}_\gamma(x_t)$ and every bad action $a^- \notin \mathcal{G}_\gamma(x_t)$, the verifier misranks the pair with probability at most η_V, i.e.,*

$$\mathbb{P}\big(V_\phi(x_t, a^-) \geq V_\phi(x_t, a^+)\big) \leq \eta_V.$$

Then the probability that verifier selection returns a γ-good action satisfies

$$\mathbb{P}\Big(\hat{a}_t^{\text{SLM}} \in \mathcal{G}_\gamma(x_t)\Big) \geq 1 - (1 - \mu_\gamma(x_t))^K - \frac{K(K-1)}{2}\eta_V.$$

Proof. We prove the bound by decomposing the failure event.

Step 1: Define the success and failure events. Let

$$E_{\text{good}} = \left\{\hat{a}_t^{\text{SLM}} \in \mathcal{G}_\gamma(x_t)\right\}.$$

We want a lower bound on $\mathbb{P}(E_{\text{good}})$.

Its complement is

$$E_{\text{bad}} = E_{\text{good}}^c = \left\{\hat{a}_t^{\text{SLM}} \notin \mathcal{G}_\gamma(x_t)\right\}.$$

Step 2: Split the bad event into two simpler events. There are only two ways selection can end up bad:

(a) No good candidate was sampled at all.

(b) At least one good candidate was sampled, but verifier misranking caused a bad candidate to be selected.

Define

$$A = \{\text{none of the } K \text{ sampled candidates lies in } \mathcal{G}_\gamma(x_t)\},$$

and

$$B = \{\text{at least one good candidate is sampled, but the selected candidate is bad}\}.$$

Then

$$E_{\text{bad}} \subseteq A \cup B.$$

Hence, by the union bound,

$$\mathbb{P}(E_{\text{bad}}) \leq \mathbb{P}(A) + \mathbb{P}(B).$$

Therefore,

$$\mathbb{P}(E_{\text{good}}) = 1 - \mathbb{P}(E_{\text{bad}}) \geq 1 - \mathbb{P}(A) - \mathbb{P}(B).$$

Step 3: Bound $\mathbb{P}(A)$. Each sampled candidate is good with probability $\mu_\gamma(x_t)$. Therefore each sampled candidate is bad with probability $1 - \mu_\gamma(x_t)$.

Because the K candidates are sampled independently,

$$\mathbb{P}(A) = (1 - \mu_\gamma(x_t))^K.$$

Step 4: Bound $\mathbb{P}(B)$. Now consider the event B: at least one good candidate exists, but a bad candidate is selected.

For this to happen, there must exist at least one good–bad pair (a^+, a^-) among the sampled candidates such that the verifier ranks the bad candidate at least as high as the good one.

For a fixed such pair, the pairwise misranking probability is at most η_V by assumption.

How many good–bad pairs can there be among K candidates? At most

$$\frac{K(K-1)}{2},$$

because this is the total number of unordered pairs among K objects.

Applying the union bound again,

$$\mathbb{P}(B) \leq \frac{K(K-1)}{2}\eta_V.$$

Step 5: Combine the two bounds. Substitute the bounds for $\mathbb{P}(A)$ and $\mathbb{P}(B)$:

$$\mathbb{P}(E_{\text{good}}) \geq 1 - (1 - \mu_\gamma(x_t))^K - \frac{K(K-1)}{2}\eta_V.$$

This proves the proposition. □

Remark A.2. *Proposition 22 explains why the verifier helps twice: first through $\mu_\gamma(x_t)$, which measures whether the SLM can* generate *at least one strong candidate, and second through η_V, which measures whether the verifier can* identify *that candidate once it appears.*

A.4 Consistency regularization under perturbation noise

Lemma 23 (Total variation is controlled by Jensen–Shannon divergence)**.** *For any two distributions P and Q,*

$$\mathrm{TV}(P, Q) \leq \sqrt{2\,\mathrm{JSD}(P\|Q)}.$$

Proof. Let

$$M = \frac{P + Q}{2}.$$

Step 1: Use the triangle inequality for total variation. By the triangle inequality,

$$\mathrm{TV}(P, Q) \le \mathrm{TV}(P, M) + \mathrm{TV}(M, Q).$$

Step 2: Apply Pinsker's inequality twice. Pinsker's inequality states that for any two distributions U, V,

$$\mathrm{TV}(U, V) \le \sqrt{\frac{1}{2}\,\mathrm{KL}(U\|V)}.$$

Apply it first with $U = P$, $V = M$:

$$\mathrm{TV}(P, M) \le \sqrt{\frac{1}{2}\,\mathrm{KL}(P\|M)}.$$

Apply it again with $U = Q$, $V = M$:

$$\mathrm{TV}(Q, M) \le \sqrt{\frac{1}{2}\,\mathrm{KL}(Q\|M)}.$$

Since $\mathrm{TV}(M, Q) = \mathrm{TV}(Q, M)$,

$$\mathrm{TV}(M, Q) \le \sqrt{\frac{1}{2}\,\mathrm{KL}(Q\|M)}.$$

Step 3: Add the two bounds. Hence

$$\mathrm{TV}(P, Q) \le \sqrt{\frac{1}{2}\,\mathrm{KL}(P\|M)} + \sqrt{\frac{1}{2}\,\mathrm{KL}(Q\|M)}.$$

Step 4: Simplify the square-root sum. For any nonnegative numbers a, b,

$$(\sqrt{a} + \sqrt{b})^2 = a + b + 2\sqrt{ab} \le 2a + 2b,$$

so

$$\sqrt{a} + \sqrt{b} \le \sqrt{2(a + b)}.$$

Apply this with

$$a = \frac{1}{2}\,\mathrm{KL}(P\|M), \qquad b = \frac{1}{2}\,\mathrm{KL}(Q\|M).$$

Then

$$\mathrm{TV}(P, Q) \le \sqrt{2\left(\frac{1}{2}\,\mathrm{KL}(P\|M) + \frac{1}{2}\,\mathrm{KL}(Q\|M)\right)}.$$

Step 5: Recognize Jensen–Shannon divergence. By definition,

$$\mathrm{JSD}(P\|Q) = \frac{1}{2}\,\mathrm{KL}(P\|M) + \frac{1}{2}\,\mathrm{KL}(Q\|M).$$

Therefore

$$\mathrm{TV}(P, Q) \le \sqrt{2\,\mathrm{JSD}(P\|Q)}.$$

This proves the lemma. □

Theorem 24 (Perturbation-generalization bound from consistency regularization)**.** *Assume Assumptions 14 and 17. For a perturbation seed z, define the episodic surrogate risk*

$$R_z(\pi_\theta) := \mathbb{E}\left[\sum_{t=1}^{H} \ell_t\Big(\pi_\theta(\cdot \mid x_t^{(z)}); s_t\Big)\right].$$

Then for any two perturbation seeds $z, z' \in \mathcal{Z}$,

$$|R_z(\pi_\theta) - R_{z'}(\pi_\theta)| \le HL\sqrt{2\,\mathcal{L}_{\mathrm{cons}}(\theta)},$$

where

$$\mathcal{L}_{\text{cons}}(\theta) = \mathbb{E}_{t,z,z'}\left[\text{JSD}\Big(\pi_\theta(\cdot \mid x_t^{(z)}) \;\Big\|\; \pi_\theta(\cdot \mid x_t^{(z')})\Big)\right].$$

Consequently, if a training seed z_{train} is representative of the seen perturbations, then for any unseen test seed z_{test},

$$R_{z_{\text{test}}}(\pi_\theta) \le R_{z_{\text{train}}}(\pi_\theta) + HL\sqrt{2\,\mathcal{L}_{\text{cons}}(\theta)}.$$

Proof. We proceed very slowly so that each transformation is explicit.

Step 1: Write the difference of the two episodic risks. By definition,

$$R_z(\pi_\theta) - R_{z'}(\pi_\theta) = \mathbb{E}\left[\sum_{t=1}^{H} \ell_t\Big(P_t^{(z)}; s_t\Big)\right] - \mathbb{E}\left[\sum_{t=1}^{H} \ell_t\Big(P_t^{(z')}; s_t\Big)\right],$$

where

$$P_t^{(z)} := \pi_\theta(\cdot \mid x_t^{(z)}), \qquad P_t^{(z')} := \pi_\theta(\cdot \mid x_t^{(z')}).$$

Using linearity of expectation and linearity of summation,

$$R_z(\pi_\theta) - R_{z'}(\pi_\theta) = \mathbb{E}\left[\sum_{t=1}^{H}\Big(\ell_t(P_t^{(z)}; s_t) - \ell_t(P_t^{(z')}; s_t)\Big)\right].$$

Step 2: Take absolute values and use the triangle inequality. We want to upper bound

$$|R_z(\pi_\theta) - R_{z'}(\pi_\theta)|.$$

Apply the triangle inequality:

$$|R_z(\pi_\theta) - R_{z'}(\pi_\theta)| \le \mathbb{E}\left[\sum_{t=1}^{H}\Big|\ell_t(P_t^{(z)}; s_t) - \ell_t(P_t^{(z')}; s_t)\Big|\right].$$

Because the sum has only nonnegative terms after the absolute value, we can pull the sum outside:

$$\le \sum_{t=1}^{H} \mathbb{E}\left[\Big|\ell_t(P_t^{(z)}; s_t) - \ell_t(P_t^{(z')}; s_t)\Big|\right].$$

Step 3: Use the Lipschitz assumption in total variation. By Assumption 17,

$$\Big|\ell_t(P_t^{(z)}; s_t) - \ell_t(P_t^{(z')}; s_t)\Big| \le L\ \text{TV}(P_t^{(z)}, P_t^{(z')}).$$

Substitute this into the previous inequality:

$$|R_z(\pi_\theta) - R_{z'}(\pi_\theta)| \le L\sum_{t=1}^{H} \mathbb{E}\Big[\text{TV}(P_t^{(z)}, P_t^{(z')})\Big].$$

Step 4: Convert total variation to JSD using Lemma 23. Lemma 23 gives

$$\text{TV}(P_t^{(z)}, P_t^{(z')}) \le \sqrt{2\ \text{JSD}(P_t^{(z)} \| P_t^{(z')})}.$$

Therefore,

$$|R_z(\pi_\theta) - R_{z'}(\pi_\theta)| \le L\sum_{t=1}^{H} \mathbb{E}\left[\sqrt{2\ \text{JSD}(P_t^{(z)} \| P_t^{(z')})}\right].$$

Step 5: Apply Jensen's inequality to move the expectation inside the square root. The function $u \mapsto \sqrt{u}$ is concave on $[0, \infty)$. For a concave function φ, Jensen's inequality states

$$\mathbb{E}[\varphi(X)] \leq \varphi(\mathbb{E}[X]).$$

Apply this with

$$\varphi(u) = \sqrt{u}, \qquad X = 2\ \mathrm{JSD}(P_t^{(z)} \| P_t^{(z')}).$$

Then

$$\mathbb{E}\left[\sqrt{2\ \mathrm{JSD}(P_t^{(z)} \| P_t^{(z')})}\right] \leq \sqrt{2\,\mathbb{E}\left[\mathrm{JSD}(P_t^{(z)} \| P_t^{(z')})\right]}.$$

So

$$|R_z(\pi_\theta) - R_{z'}(\pi_\theta)| \leq L \sum_{t=1}^{H} \sqrt{2\,\mathbb{E}\left[\mathrm{JSD}(P_t^{(z)} \| P_t^{(z')})\right]}.$$

Step 6: Use the global consistency regularizer to upper bound each step. By definition,

$$\mathcal{L}_{\mathrm{cons}}(\theta) = \mathbb{E}_{t,z,z'}\left[\mathrm{JSD}(P_t^{(z)} \| P_t^{(z')})\right].$$

In particular, for the pair (z, z') being considered, the average stepwise JSD is upper bounded by $\mathcal{L}_{\mathrm{cons}}(\theta)$ up to the natural empirical averaging interpretation. Using this quantity as the global bound,

$$\mathbb{E}\left[\mathrm{JSD}(P_t^{(z)} \| P_t^{(z')})\right] \leq \mathcal{L}_{\mathrm{cons}}(\theta)$$

for each t in the worst case or on average over the paired dataset.

Thus each summand is at most

$$\sqrt{2\,\mathcal{L}_{\mathrm{cons}}(\theta)}.$$

There are H such summands, so

$$|R_z(\pi_\theta) - R_{z'}(\pi_\theta)| \leq HL\sqrt{2\,\mathcal{L}_{\mathrm{cons}}(\theta)}.$$

This proves the first inequality.

Step 7: Deduce the train-to-test bound. Take $z = z_{\mathrm{test}}$ and $z' = z_{\mathrm{train}}$. Then

$$R_{z_{\mathrm{test}}}(\pi_\theta) - R_{z_{\mathrm{train}}}(\pi_\theta) \leq |R_{z_{\mathrm{test}}}(\pi_\theta) - R_{z_{\mathrm{train}}}(\pi_\theta)| \leq HL\sqrt{2\,\mathcal{L}_{\mathrm{cons}}(\theta)}.$$

Rearranging yields

$$R_{z_{\mathrm{test}}}(\pi_\theta) \leq R_{z_{\mathrm{train}}}(\pi_\theta) + HL\sqrt{2\,\mathcal{L}_{\mathrm{cons}}(\theta)}.$$

This proves the theorem. □

Remark A.3. *Theorem 24 is written for an episodic* surrogate *risk that decomposes over steps. If the binary episode failure $1 - S_z$ is upper bounded by such a surrogate, then the same bound transfers immediately to the failure objective used in the robust routing section.*

A.5 CVaR-constrained primal–dual optimization

Definition A.2 (Empirical CVaR over perturbation seeds)**.** *For each perturbation seed $z \in \mathcal{Z}$, let the empirical failure rate be*

$$F_z = 1 - \mathrm{SR}_z.$$

Let

$$F_{(1)} \geq F_{(2)} \geq \cdots \geq F_{(|\mathcal{Z}|)}$$

denote the sorted seed-wise failure rates, and let

$$k = \lceil \alpha |\mathcal{Z}| \rceil.$$

Then

$$\widehat{\mathrm{CVaR}}_\alpha(1 - S_z) = \frac{1}{k} \sum_{j=1}^{k} F_{(j)}.$$

Under the convex surrogate in Assumption 18, consider the empirical Lagrangian

$$\widehat{\mathcal{L}}_{\text{router}}(\psi, \lambda) = \widehat{\mathbb{E}}[\text{cost}(d)] + \lambda\Big(\widehat{\text{CVaR}}_\alpha(1 - S_z) - \varepsilon\Big) + \widehat{\text{Brier}}(r_\psi(f), y).$$

We run projected gradient descent–ascent:

$$\psi_{n+1} = \Pi_\Psi\Big(\psi_n - \eta\nabla_\psi\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda_n)\Big),$$
$$\lambda_{n+1} = \Pi_{[0,\Lambda]}\Big(\lambda_n + \eta\nabla_\lambda\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda_n)\Big),$$

and define the averaged iterates

$$\bar{\psi}_N = \frac{1}{N}\sum_{n=1}^{N}\psi_n, \qquad \bar{\lambda}_N = \frac{1}{N}\sum_{n=1}^{N}\lambda_n.$$

Theorem 25 (Primal–dual convergence and near-feasibility)**.** *Assume Assumptions 14 and 18. Let the diameter of Ψ be at most D_ψ, and the diameter of $[0, \Lambda]$ be at most $D_\lambda = \Lambda$. Define*

$$D := \sqrt{D_\psi^2 + D_\lambda^2}.$$

If the step size is chosen as

$$\eta = \frac{D}{G\sqrt{N}},$$

then the averaged iterates satisfy the saddle-point gap bound

$$\max_{\lambda\in[0,\Lambda]}\widehat{\mathcal{L}}_{\text{router}}(\bar{\psi}_N, \lambda) - \min_{\psi\in\Psi}\widehat{\mathcal{L}}_{\text{router}}(\psi, \bar{\lambda}_N) \le \frac{2DG}{\sqrt{N}}.$$

Moreover, the empirical CVaR-constraint violation satisfies

$$\Big[\widehat{\text{CVaR}}_\alpha(1 - S_z; \bar{\psi}_N) - \varepsilon\Big]_+ \le \frac{2DG}{\Lambda\sqrt{N}}.$$

Proof. The proof is a standard but important telescoping argument. I spell out each algebraic step.

Step 1: Introduce shorthand for the gradients. Define

$$g_n^\psi := \nabla_\psi\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda_n), \qquad g_n^\lambda := \nabla_\lambda\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda_n).$$

By assumption,

$$\|g_n^\psi\|_2 \le G, \qquad |g_n^\lambda| \le G.$$

Step 2: Use nonexpansiveness of projection for the primal update. Fix any comparator $\psi \in \Psi$. Because Euclidean projection onto a convex set is nonexpansive,

$$\|\psi_{n+1} - \psi\|_2^2 \le \|\psi_n - \eta g_n^\psi - \psi\|_2^2.$$

Expand the square on the right-hand side:

$$\|\psi_n - \psi\|_2^2 - 2\eta\langle g_n^\psi, \psi_n - \psi\rangle + \eta^2\|g_n^\psi\|_2^2.$$

Therefore,

$$\|\psi_{n+1} - \psi\|_2^2 \le \|\psi_n - \psi\|_2^2 - 2\eta\langle g_n^\psi, \psi_n - \psi\rangle + \eta^2 G^2.$$

Rearrange:

$$2\eta\langle g_n^\psi, \psi_n - \psi\rangle \le \|\psi_n - \psi\|_2^2 - \|\psi_{n+1} - \psi\|_2^2 + \eta^2 G^2.$$

Divide by 2η:

$$\langle g_n^\psi, \psi_n - \psi\rangle \le \frac{\|\psi_n - \psi\|_2^2 - \|\psi_{n+1} - \psi\|_2^2}{2\eta} + \frac{\eta G^2}{2}.$$

Step 3: Convert the inner product into a primal function gap using convexity. Because the Lagrangian is convex in ψ,

$$\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda_n) - \widehat{\mathcal{L}}_{\text{router}}(\psi, \lambda_n) \leq \langle g_n^{\psi}, \psi_n - \psi \rangle.$$

Combining with the previous inequality,

$$\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda_n) - \widehat{\mathcal{L}}_{\text{router}}(\psi, \lambda_n) \leq \frac{\|\psi_n - \psi\|_2^2 - \|\psi_{n+1} - \psi\|_2^2}{2\eta} + \frac{\eta G^2}{2}.$$

Step 4: Do the same for the dual variable. Now fix any comparator $\lambda \in [0, \Lambda]$.

The dual update is an ascent step followed by projection:

$$\lambda_{n+1} = \Pi_{[0,\Lambda]}(\lambda_n + \eta g_n^{\lambda}).$$

Again by nonexpansiveness of projection,

$$|\lambda_{n+1} - \lambda|^2 \leq |\lambda_n + \eta g_n^{\lambda} - \lambda|^2.$$

Expand:

$$|\lambda_n - \lambda|^2 + 2\eta g_n^{\lambda}(\lambda_n - \lambda) + \eta^2 |g_n^{\lambda}|^2.$$

So

$$|\lambda_{n+1} - \lambda|^2 \leq |\lambda_n - \lambda|^2 + 2\eta g_n^{\lambda}(\lambda_n - \lambda) + \eta^2 G^2.$$

Rearrange:

$$2\eta g_n^{\lambda}(\lambda - \lambda_n) \leq |\lambda_n - \lambda|^2 - |\lambda_{n+1} - \lambda|^2 + \eta^2 G^2.$$

Divide by 2η:

$$g_n^{\lambda}(\lambda - \lambda_n) \leq \frac{|\lambda_n - \lambda|^2 - |\lambda_{n+1} - \lambda|^2}{2\eta} + \frac{\eta G^2}{2}.$$

Step 5: Convert the dual inner product into a function gap using concavity. Because the Lagrangian is concave in λ,

$$\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda) - \widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda_n) \leq g_n^{\lambda}(\lambda - \lambda_n).$$

Therefore,

$$\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda) - \widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda_n) \leq \frac{|\lambda_n - \lambda|^2 - |\lambda_{n+1} - \lambda|^2}{2\eta} + \frac{\eta G^2}{2}.$$

Step 6: Add the primal and dual inequalities. Add the inequalities from Step 3 and Step 5:

$$\begin{aligned}
&\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda) - \widehat{\mathcal{L}}_{\text{router}}(\psi, \lambda_n) \\
&= \left[\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda) - \widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda_n)\right] + \left[\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda_n) - \widehat{\mathcal{L}}_{\text{router}}(\psi, \lambda_n)\right] \\
&\leq \frac{\|\psi_n - \psi\|_2^2 - \|\psi_{n+1} - \psi\|_2^2}{2\eta} + \frac{|\lambda_n - \lambda|^2 - |\lambda_{n+1} - \lambda|^2}{2\eta} + \eta G^2.
\end{aligned}$$

Step 7: Sum from $n = 1$ to N. Summing both sides,

$$\begin{aligned}
&\sum_{n=1}^{N} \left(\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda) - \widehat{\mathcal{L}}_{\text{router}}(\psi, \lambda_n)\right) \\
&\leq \frac{\|\psi_1 - \psi\|_2^2 - \|\psi_{N+1} - \psi\|_2^2}{2\eta} + \frac{|\lambda_1 - \lambda|^2 - |\lambda_{N+1} - \lambda|^2}{2\eta} + N\eta G^2.
\end{aligned}$$

Since squared norms are nonnegative, we may drop the negative terminal terms:

$$\leq \frac{\|\psi_1 - \psi\|_2^2}{2\eta} + \frac{|\lambda_1 - \lambda|^2}{2\eta} + N\eta G^2.$$

By the diameter bounds,

$$\|\psi_1 - \psi\|_2 \leq D_\psi, \qquad |\lambda_1 - \lambda| \leq D_\lambda.$$

So

$$\sum_{n=1}^{N} \Big(\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda) - \widehat{\mathcal{L}}_{\text{router}}(\psi, \lambda_n) \Big) \leq \frac{D_\psi^2 + D_\lambda^2}{2\eta} + N\eta G^2 = \frac{D^2}{2\eta} + N\eta G^2.$$

Step 8: Divide by N and move to the averaged iterates. Divide by N:

$$\frac{1}{N} \sum_{n=1}^{N} \Big(\widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda) - \widehat{\mathcal{L}}_{\text{router}}(\psi, \lambda_n) \Big) \leq \frac{D^2}{2\eta N} + \eta G^2.$$

By convexity in ψ and concavity in λ,

$$\widehat{\mathcal{L}}_{\text{router}}(\bar{\psi}_N, \lambda) \leq \frac{1}{N} \sum_{n=1}^{N} \widehat{\mathcal{L}}_{\text{router}}(\psi_n, \lambda),$$

and

$$\widehat{\mathcal{L}}_{\text{router}}(\psi, \bar{\lambda}_N) \geq \frac{1}{N} \sum_{n=1}^{N} \widehat{\mathcal{L}}_{\text{router}}(\psi, \lambda_n).$$

Subtracting the two inequalities gives

$$\widehat{\mathcal{L}}_{\text{router}}(\bar{\psi}_N, \lambda) - \widehat{\mathcal{L}}_{\text{router}}(\psi, \bar{\lambda}_N) \leq \frac{D^2}{2\eta N} + \eta G^2.$$

Step 9: Choose the step size. Set

$$\eta = \frac{D}{G\sqrt{N}}.$$

Then

$$\frac{D^2}{2\eta N} = \frac{D^2}{2(D/(G\sqrt{N}))N} = \frac{DG}{2\sqrt{N}},$$

and

$$\eta G^2 = \frac{D}{G\sqrt{N}} G^2 = \frac{DG}{\sqrt{N}}.$$

Therefore

$$\widehat{\mathcal{L}}_{\text{router}}(\bar{\psi}_N, \lambda) - \widehat{\mathcal{L}}_{\text{router}}(\psi, \bar{\lambda}_N) \leq \frac{3DG}{2\sqrt{N}}.$$

For cleanliness, we can upper bound this by

$$\frac{2DG}{\sqrt{N}}.$$

Because the inequality holds for all $\psi \in \Psi$ and $\lambda \in [0, \Lambda]$,

$$\max_{\lambda \in [0,\Lambda]} \widehat{\mathcal{L}}_{\text{router}}(\bar{\psi}_N, \lambda) - \min_{\psi \in \Psi} \widehat{\mathcal{L}}_{\text{router}}(\psi, \bar{\lambda}_N) \leq \frac{2DG}{\sqrt{N}}.$$

This proves the saddle-gap bound.

Step 10: Convert saddle gap into a constraint-violation bound. Now define

$$g(\psi) = \widehat{\mathrm{CVaR}}_\alpha(1 - S_z; \psi) - \varepsilon.$$

For fixed $\bar{\psi}_N$, the Lagrangian is affine in λ:

$$\widehat{\mathcal{L}}_{\text{router}}(\bar{\psi}_N, \lambda) = \widehat{\mathbb{E}}[\text{cost}(d)] + \widehat{\text{Brier}}(r_{\bar{\psi}_N}(f), y) + \lambda g(\bar{\psi}_N).$$

If $g(\bar{\psi}_N) \leq 0$, then the positive part $[g(\bar{\psi}_N)]_+ = 0$, and the desired bound is trivial.

If $g(\bar{\psi}_N) > 0$, then maximizing over $\lambda \in [0, \Lambda]$ sets $\lambda = \Lambda$, so

$$\max_{\lambda \in [0,\Lambda]} \widehat{\mathcal{L}}_{\text{router}}(\bar{\psi}_N, \lambda) = \widehat{\mathbb{E}}[\text{cost}(d)] + \widehat{\text{Brier}}(r_{\bar{\psi}_N}(f), y) + \Lambda g(\bar{\psi}_N).$$

In both cases,

$$\max_{\lambda \in [0,\Lambda]} \widehat{\mathcal{L}}_{\text{router}}(\bar{\psi}_N, \lambda) = \widehat{\mathbb{E}}[\text{cost}(d)] + \widehat{\text{Brier}}(r_{\bar{\psi}_N}(f), y) + \Lambda [g(\bar{\psi}_N)]_+.$$

Subtract

$$\min_{\psi \in \Psi} \widehat{\mathcal{L}}_{\text{router}}(\psi, \bar{\lambda}_N)$$

from both sides and use the saddle-gap bound:

$$\Lambda [g(\bar{\psi}_N)]_+ \leq \frac{2DG}{\sqrt{N}}.$$

Finally divide by Λ:

$$[g(\bar{\psi}_N)]_+ \leq \frac{2DG}{\Lambda\sqrt{N}}.$$

Substituting the definition of g,

$$\left[\widehat{\mathrm{CVaR}}_\alpha(1 - S_z; \bar{\psi}_N) - \varepsilon\right]_+ \leq \frac{2DG}{\Lambda\sqrt{N}}.$$

This proves the theorem. □

A.6 Optional appendix result: noisy-verifier DPO consistency

Proposition 26 (Pairwise sign consistency of verifier-noisy DPO)**.** *Consider a fixed context x_t and two actions a^+, a^-. Let the true pairwise preference label be*

$$y^\star \in \{+1, -1\},$$

where $y^\star = +1$ means a^+ is truly preferred to a^-. Suppose the verifier produces a noisy pairwise label $\widetilde{y}$ such that

$$\mathbb{P}(\widetilde{y} = y^\star \mid x_t, a^+, a^-) = 1 - \eta, \qquad \mathbb{P}(\widetilde{y} = -y^\star \mid x_t, a^+, a^-) = \eta,$$

with $\eta < 1/2$.

Let the DPO score difference be

$$u_\theta(x_t, a^+, a^-) = \beta \left[\log \frac{\pi_\theta(a^+ \mid x_t)}{\pi_{\text{ref}}(a^+ \mid x_t)} - \log \frac{\pi_\theta(a^- \mid x_t)}{\pi_{\text{ref}}(a^- \mid x_t)}\right].$$

Then the population minimizer of the noisy logistic loss

$$\mathcal{L}_{\text{DPO,noisy}}(\theta) = \mathbb{E}\Big[\log(1 + \exp(-\widetilde{y}\, u_\theta))\Big]$$

has the same sign as the true preference:

$$\text{sign}\big(u_\theta^\star(x_t, a^+, a^-)\big) = y^\star.$$

Proof. We prove the result for $y^\star = +1$. The case $y^\star = -1$ is symmetric.

Conditional on the pair (x_t, a^+, a^-), the noisy label satisfies

$$\widetilde{y} = \begin{cases} +1, & \text{with probability } 1-\eta, \\ -1, & \text{with probability } \eta. \end{cases}$$

Therefore, as a function of the scalar score u, the conditional risk is

$$R(u) = (1-\eta)\log(1+e^{-u}) + \eta \log(1+e^{u}).$$

Differentiate:

$$R'(u) = -(1-\eta)\frac{1}{1+e^u} + \eta \frac{e^u}{1+e^u}.$$

Put both terms over the common denominator $1+e^u$:

$$R'(u) = \frac{\eta e^u - (1-\eta)}{1+e^u}.$$

Set the derivative to zero:

$$\eta e^u - (1-\eta) = 0 \quad \Longleftrightarrow \quad e^u = \frac{1-\eta}{\eta}.$$

Since $\eta < 1/2$, we have

$$\frac{1-\eta}{\eta} > 1,$$

hence

$$u^\star = \log\left(\frac{1-\eta}{\eta}\right) > 0.$$

So the minimizing score is positive whenever the true preference is positive.

By symmetry, if $y^\star = -1$, then the minimizing score is negative.

Thus the optimal noisy-DPO score still has the correct sign as long as $\eta < 1/2$.

This proves the proposition. □